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A semi-supervised deep learning based video anomaly detection framework using RGB-D for surveillance of real-world critical environments

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ABSTRACT

Video surveillance has become very important in the current era of smart cities. Large amounts of surveillance cameras are deployed at public and private places for surveillance of infrastructural property and public safety. These surveillance cameras generate huge amount of video data and it is impractical for a human observer to continuously monitor these long-hour videos manually and detect any unwanted or anomalous event. This paper presents a multi-modal semi-supervised deep learning based CNN-BiLSTM autoencoder framework to detect anomalous events in critical surveillance environments like Bank-ATMs. The significance of the framework is that it only requires weakly labelled normal video samples for training. We leverage the power of transfer learning by extracting important video features using a compact pretrained CNN to significantly reduce the computational complexity of training and detection. Moreover, due to the unavailability of any dataset for ATM surveillance in the public domain, we also contributed a unique RGB + D dataset for surveillance of ATMs. The proposed framework is tested on the collected RGB + D dataset and other real-world benchmark video anomaly datasets: Avenue and UCFCrime2Local. Results show that the proposed framework gives competitive results with other state-of-the-art methods and can be applied to both indoor and outdoor environments for detection of anomalies in real-world surveillance sites.

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1. Introduction

Video surveillance has become very important and gaining increasing popularity in the modern days of smart city era. An enormous amount of surveillance cameras are installed at public and private places like Bank-ATMs (Automatic Teller Machines), shopping malls, traffic signals, offices, railway platforms, homes, academic environments, etc. for surveillance of infrastructural property and public safety (Nayak et al., 2020; Ribeiro et al., 2018). These surveillance cameras continuously generate large amount of video data. It is very difficult and time-consuming task for a human observer to manually monitor these long-hour live video streams and detect the presence of any abnormal event or unwanted incident (Nayak et al., 2020). Automatic detection of such anomalous events in videos can significantly reduce human effort in monitoring. Therefore, video anomaly detection is an important research problem in the area of computer vision (Ribeiro et al., 2018).

Video anomaly detection is defined as the process of automatic detection of abnormal events in videos through video processing and computer vision techniques. The term 'anomaly', 'abnormal', 'unusual', 'outliers' etc. are often used to refer the irregular pattern in videos that is different from the regular one (Nayak et al., 2020). The anomalous or unusual events are often subjective, contextual in nature, and environment-dependent (Nayak et al., 2020; Ribeiro et al., 2018). Unwanted incidents such as fire explosions, stampede, accidents in public places, and irregular human activities like fighting, robbery, violations of traffic rules are some of the challenging tasks in video anomaly detection (Gowsikhaa et al., 2014). Detecting such anomalous events at public places can prevent infrastructural damages and save lives of people. Hence, automatic and accurate detection of such events in real-time is the main objective of video-based surveillance systems.

The readers must note that, although video anomaly detection and video anomaly localization are interrelated concepts with significant difference (Nayak et al., 2020), anomaly detection is the

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process of detecting the anomaly in the video or video clips/frames while anomaly localization is the process of localizing the detected anomaly in the frame at pixel level (Pawar and Attar, 2019). For detailed discussion, readers are referred to some of the noted survey papers on video based anomaly detection in (Nayak et al., 2020; Ribeiro et al., 2018; Gowsikhaa et al., 2014; Pawar and Attar, 2019; Tripathi et al., 2018). The foremost challenge in video anomaly detection is the less availability or unavailability of the anomalous video samples, as the anomalies are the abnormal events that occurs rarely (Ribeiro et al., 2018; Pawar and Attar, 2019). Moreover, the high variations among the anomalies result in an unbalanced data for modeling and training. For supervised learning, the training data must be clearly defined with the associated labels.

In recent years, with the success of deep learning on image classification (Krizhevsky et al., 2012; Simonyan and Zisserman, 2014) and object detection tasks (Girshick et al., 2014; Ren et al., 2016), deep learning-based supervised, unsupervised, and semi-supervised techniques are also gaining popularity for video anomaly detection. Many of the researchers have used supervised learning methods (Zhang et al., 2017; Singh and Pankajakshan, 2018; Sultani et al., 2018; Gianchandani et al., 2019; Landi et al., 2019; Parab et al., 2020; Ullah et al., 2020) for video anomaly detection. However, the major disadvantage of these approaches is that they require balanced and labeled data of normal and anomalous samples for supervised training. Hence, the supervised learning models are not the best-suited choice for video anomaly detection (Nayak et al., 2020).

Deep learning based semi-supervised methods for video anomaly detection can be found in (Sabokrou et al., 2017; Luo et al., 2017; Ribeiro et al., 2018; Bhakat and Ramakrishnan, 2019; Park et al., 2020). The significance of semi-supervised and unsupervised methods is that they can detect the anomalies using the weakly labeled or unlabeled data, where only the normal video samples are given as input for training (Sabokrou et al., 2017; Bhakat and Ramakrishnan, 2019). As reviewed by (Ribeiro et al., 2018), most of the approaches are based on deep autoencoders that utilizes the raw video frames of normal data as input to the encoder and reconstruct the input frames at the decoder output by learning the normal pattern of frame sequences. However, a video contains spatial redundancy among pixels in both foreground and background in a video frame and temporal redundancy among consecutive video frames. Furthermore, anomalous events occur rarely both in the spatial and temporal domain in a video (Ribeiro et al., 2018; Park et al., 2020). Therefore, the computational overhead of reconstructing each raw video frame both spatially and temporally at the decoder output is the major drawback of these deep learning based autoencoder methods.

There are various other challenges to video anomaly detection including variations in illuminations, complex background, object occlusions, presence of noise in data, etc. that makes it one of the difficult tasks in computer vision (Nayak et al., 2020; Ribeiro et al., 2018; Pawar and Attar, 2019). In recent years, with the availability of depth sensors such as Microsoft Kinect (Zhang, 2012) and Intel real sense (Mejia-Trujillo et al., 2019) at affordable costs, research in computer vision is advancing towards the use of these sensors. The (Red, Green, Blue) RGB + D (Depth) sensor provides depth data of the scene in addition to (Red, Green, Blue) RGB data, where the depth data is less sensitive to illumination variations and is invariant to color and texture changes (Giancola et al., 2018). Numerous RGB + D datasets (Zhang et al., 2016; Sung et al., 2012; Koppula et al., 2013; Chen et al., 2015; Shahroudy et al., 2016) have been collected in recent years for human-related activities in the office, kitchen environment, etc. Further details on RGB + D datasets can be found in the survey papers (Zhang et al., 2016; Jegham et al., 2020).

In this paper, to address the problem of anomaly detection in critical environments such as Bank-ATMs, we propose to use multi-modal RGB and depth data captured by advanced RGB + D sensor. By observing the human behaviour in ATM and the activities that frequently occur, we replicated and captured various normal and anomalous activities in ATM like environment to create a dataset. Fig. 1 shows the sample normal and anomalous human activities captured using the RGB + D sensor.

Following are the major contributions/salient features of the paper:

1. This paper presents a deep learning based semi-supervised (Convolutional Neural Networks - Bidirectional Long Short Term Memory) CNN-BiLSTM autoencoder framework using multi-modal RGB + D data for real-time online video anomaly detection in surveillance videos. To the best of our knowledge, this is the first semi-supervised deep learning framework using RGB+D data for video anomaly detection in critical environments such as Bank-ATMs.
2. The proposed CNN-BiLSTM autoencoder framework is semi-supervised and requires only weakly labeled video samples for training. Moreover, it does not require any anomalous video samples for training to detect anomalies. Hence, the proposed framework overcomes the limitations of training the unbalanced normal and anomalous data.
3. Furthermore, due to the unavailability of any dataset for surveillance of Bank-ATM in the public domain. We also contributed a unique multi-modal RGB+D (RGB, Depth, Skeleton) dataset for testing and evaluation of anomaly detection methods in Bank-ATM environment.
4. The proposed framework is evaluated on other real-world benchmark datasets: Avenue (Lu et al., 2013) and UCFCrime2-Local (Landi et al., 2019) collected in different environments and the results obtained are competitive with other state-of-the-art approaches. The experimentation show that it can be applied to both indoor and outdoor environments for the detection of anomalies in real-world surveillance sites.

The rest of the paper is organized as follows: Section 2 briefly discusses the related work on video anomaly detection in Bank-ATMs and other real-world surveillance environments. Section 3 presents the proposed semi-supervised CNN-BiLSTM framework for real-time video anomaly detection. Section 4 discusses the significance of the collected RGB + D dataset for ATM surveillance with detailed statistics. Results of the proposed approach with quantitative and qualitative evaluation parameters are discussed in section 5. The conclusion and future work are discussed in Section 6.

2. Related work

This section discusses the methods for video anomaly detection in public environments. The discussion starts with the review on video-based anomaly detection methods for surveillance of Bank-ATMs followed by the review of approaches applied in other public environments.

Few works related to detection of anomalous/abnormal behaviour in ATM can be found in (Liu et al., 2012; Nar et al., 2016; Lee et al., 2018). Liu et al. (2012), detect abnormal human behaviors of peeping and wandering using RGB + D camera. Nar et al. (2016) detect three abnormal behaviors of aggressive, fiddling, and peeping. Similarly Lee et al. (2018), detect the squat and climb action using the background subtraction method. However, these approaches are heuristic-based and there are no data/datasets, quantitative results that are reported in these works. The main

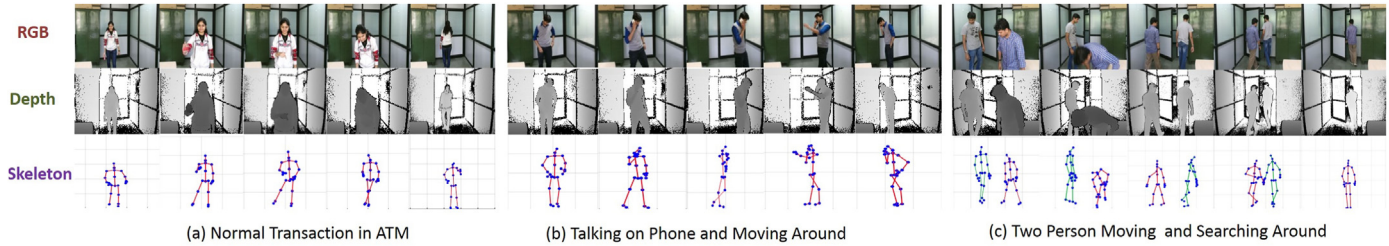

Fig. 1. Sample RGB, depth and skeletal frames for normal and anomalous activities that occurs in an ATM environment. (a) Shows the normal activity of transaction in ATM (b) Shows the anomalous activity of taking on cell phone (c) Shows the anomalous activity of two person moving and searching around in ATM.

limitations of the reported techniques are that they are handcrafted feature-based heuristic methods applied to distinguish only two-three activities in a given ATM environment which is not sufficient to analyze the robustness of the work. In general, handcrafted features-based heuristic methods often work only in the given environmental settings and may not work in other similar real-world environments (Nayak et al., 2020; Ribeiro et al., 2018). Moreover, the absence of datasets and quantitative/qualitative results without any statistical evaluation parameters reported in these works limits their utility and applicability. The following paragraph discusses video anomaly detection methods in other public environments.

Many researchers have used deep learning methods for video anomaly detection based on supervised learning (Sabokrou et al., 2017; Ullah et al., 2020; Sultani et al., 2018; Smeureanu et al., 2017; Gianchandani et al., 2019; Landi et al., 2019; Singh and Pankajakshan, 2018). For instance, Sultani et al. (2018) detected the anomalies using a deep multi-instance ranking technique. Here, in the first stage video segments are created from the video stream and the (3-dimensional) 3D features are extracted from these video segments using pre-trained 3D CNNs. The multi-instance ranking scheme is then applied to the extracted features for classifying the normal and anomalous samples in a supervised manner. Ullah et al. (2020) applied a CNN-LSTM based supervised framework where the features are extracted from normal and anomalous video samples through pre-trained (2-dimensional) 2D-CNN (Convolutional Neural Networks) ResNet (He et al., 2016) and the extracted features are given to LSTM for learning the motion information in segments. However, the disadvantage of these approaches is that they are supervised learning-based, where large number of anomalous samples are required for training.

Several semi-supervised and unsupervised works for anomaly detection can be found in (Hasan et al., 2016; Zhao et al., 2017; Luo et al., 2017; Ribeiro et al., 2018; Chong and Tay, 2017; Tang et al., 2020; Park et al., 2020; Lee et al., 2020). Most of the works are based on CNN autoencoder for learning the normal representation of frames. The significance of the auto-encoders is that it only requires weakly labeled normal samples for anomaly detection. The autoencoder takes the video frames of normal data as input and reconstructs the frames at the output by learning the normal pattern (Ribeiro et al., 2018; Hasan et al., 2016). For instance, a fully convolutional autoencoder-decoder is trained by Hasan et al. (2016), where the raw color input frame cuboid is given as input to the autoencoder. Here, the color frames are stacked together to create a temporal cuboid and then given to the 2D CNN encoder as input. Finally, the temporal cuboid of the raw frames is reconstructed at the decoder output. Zhao et al. (2017) argued that the stacking of T frames in 3rd dimension loses the temporal information. To solve this problem, they created a hyper-cuboid by stacking the frames in 4th dimension and created a 3D CNN based autoencoder-decoder framework. However, the process of reconstructing cuboids with raw color frames of large size is computationally expensive.

Luo et al. (2017) reported that 3D convolutions do not capture the temporal motion information across frames accurately and therefore created (Convolutional LSTM) ConvLSTM based autoencoder framework for appropriate modeling of temporal information across frames. The framework consists of a hierarchical layer of CNNs and ConvLSTM, where the encoder takes the raw input frames as the input and creates the raw frames at the output of the decoder. Similarly, Chong and Tay (2017) created a hierarchical spatio-temporal autoencoder by applying the CNN and ConvLSTM for spatial and temporal modeling of frame sequences. The spatial encoder encodes the spatial structure of the input frame and the temporal encoder learns the temporal information of the spatially encoded structures. The decoder reconstructs the spatio-temporal frames. Similarly, Ribeiro et al. (2018) proposed a convolutional auto-encoder decoder architecture where it takes the raw input frames, edge map, and optical flow of frames as an input to the autoencoder. The optical maps of consecutive frames are constructed to capture the motion information across frames. However, the reconstruction of raw input frames along with edge map and optical flow is a computationally expensive procedure.

In summary, supervised approaches (Sabokrou et al., 2017; Ullah et al., 2020; Sultani et al., 2018; Smeureanu et al., 2017; Gianchandani et al., 2019; Landi et al., 2019; Singh and Pankajakshan, 2018) of video anomaly detection require labeled data of normal and abnormal samples which is the major drawback of these methods. Whereas, semi-supervised based autoencoder approaches (Hasan et al., 2016; Zhao et al., 2017; Luo et al., 2017; Ribeiro et al., 2018; Chong and Tay, 2017; Tang et al., 2020; Park et al., 2020; Lee et al., 2020) reviewed in the previous paragraph have the computational overhead of reconstructing raw video frames both spatially and temporally. To address these challenges of video anomaly detection in real-world environments, we proposed a semi-supervised deep learning framework for real-time video anomaly detection in critical environments like Bank-ATMs.

3. Proposed framework

This section discusses the proposed multi-modal semi-supervised deep learning framework for video anomaly detection in Bank-ATM environment. From online video stream of RGB+D sensor, video segments are created by dividing the video stream into fixed-size video frames. Each video segment is then given to semi-supervised video anomaly detection framework to detect the presence of any anomalous activity or unwanted incident. On detection of any anomalous activity or unwanted incident, an alarm can be raised to the concerned security personal/authorities for threat detection and corrective action.

Fig. 2 shows the proposed semi-supervised CNN−Bi-LSTM encoder-decoder framework. The framework consists of two parts (a) Feature extraction module consisting of pre-trained CNN MobileNet and (b) Bi-LSTM encoder-decoder module. As shown in

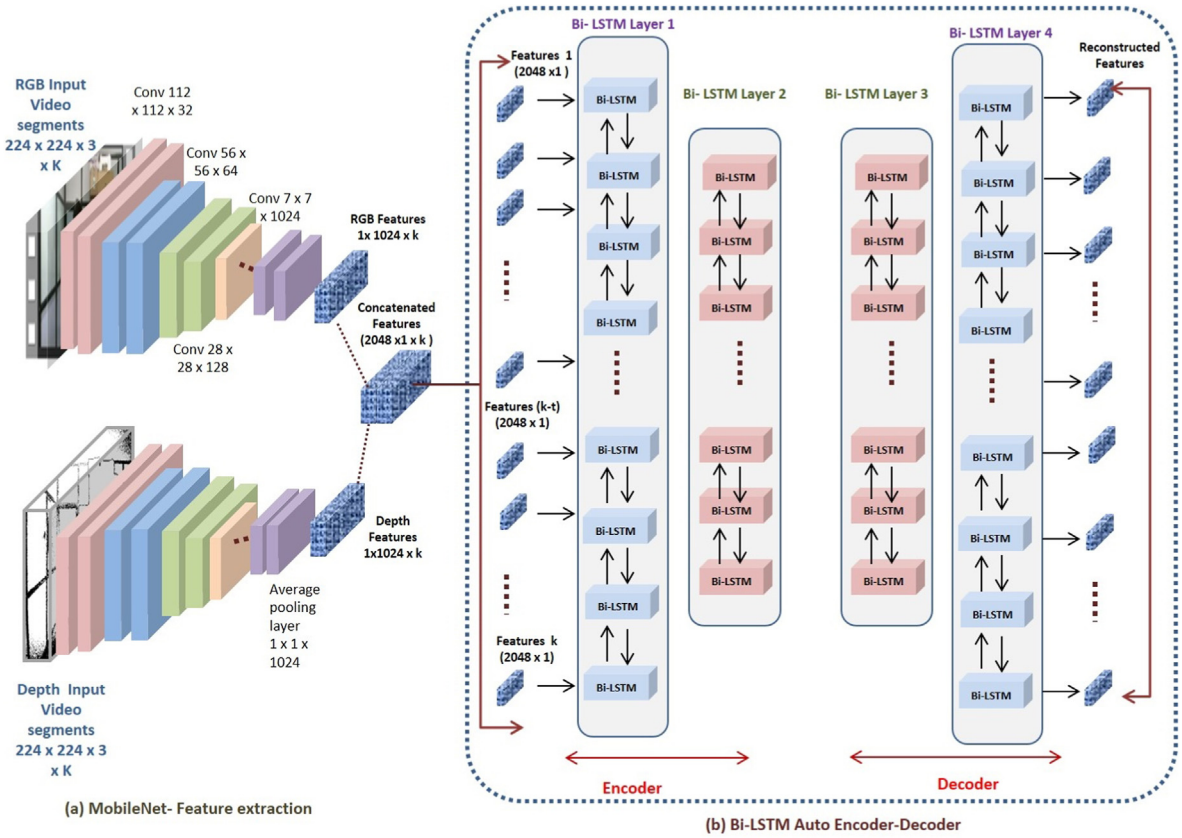

Fig. 2. Semi-supervised end-to-end framework for abnormal activity recognition. (a) Feature extraction module: Extract the features from raw RGB and depth video frames. (b) Bidirectional-LSTM Encoder-Decoder: To learn the combine features for normal activity pattern.

figure, RGB and Depth video segments are given as input to the pre-trained MobileNet model for features extraction. The extracted RGB and Depth features are then concatenated with each other and given to Bi-LSTM autoencoder to learn the normal pattern sequences. For the training purpose, only normal samples without anomalies are given to the autoencoder to learn the spatiotemporal pattern for normal scenarios. The Encoder-decoder architecture learns the normal temporal sequence of spatially extracted combined (RGB+Depth) feature segments. Once the autoencoder is trained for normal pattern, then during testing of the online video stream, the input video segment samples are given to the autoencoder to reconstruct the video segment features.

As the autoencoder-decoder is trained for only the normal pattern of video feature segments, it will try to reconstruct the normal spatio-temporal feature segment patterns using the given input sequence. The reconstruction error between the input feature segment and the reconstructed output feature segment is calculated. As the model is trained only for the normal behavior of spatiotemporal feature segments, video feature segments with abnormal or anomalous activities will have a high reconstruction error. If the error is beyond a certain threshold, then the given video segment can be recognized as the abnormal video sequence which deviates from the normal ones. The next subsection describes feature extraction module and Bi-LSTM encoder-decoder module in detail.

3.1. Feature extraction

As discussed in the related work section, the encoding and reconstruction of the raw frames at decoder output for training the normal spatio-temporal pattern is computationally expensive. For instance, consider a video V (H, W, N) of large size where, H and W denote the spatial height and width of the video frame and N are the number of frames, contains an anomalous event temporally located at the last 10 frames in a video at time t (N-10) to t+10 (N) frames. Then, for the detection of the anomalous event, the spatial and temporal frames of size (HxWxN) are reconstructed at the decoder output which is computationally expensive. Furthermore, a video frame consists of spatial redundant pixels which are less informative and contains the temporal redundancy across consecutive frames; therefore the reconstruction of the raw frames has the overhead of reconstructing the redundant information both spatially and temporally.

To overcome this problem, we leverage the power of transfer learning by extracting important video features using a compact pre-trained CNN model to significantly reduce the computational complexity of training and detection. Here, to extract the rich visual features from raw video frames, a pre-trained compact CNN model MobileNet (Howard et al., 2017) is applied in the proposed framework. The advantage of using the pre-trained MobileNet model is that it is very compact with a size of 8 MB and designed for image classification and object detection applications for Mobile and other embedded devices with less computational resources. Therefore, this feature extraction module can be easily implemented in embedded devices at the ATM surveillance sites with fewer computational requirements.

From each frame of video segments, spatial visual features are extracted through pre-trained MobileNet model. Fig. 2 (a) shows the feature extraction process from RGB and depth video segments. The features are extracted from the global average pooling layer of

the MobileNet of size (1024 × 1 x 1). Fig. 3 shows the layered architecture of the pre-trained CNN MobileNet model. The details of the MobileNet architecture and layers with its description can be found in (Howard et al., 2017). Here, the readers must note that the features are extracted from the global pooling layer after the hierarchy of convolution layers and not from the fully connected layer of classification. As analyzed and reported in (Long et al., 2015; Hasan et al., 2016), a fully connected layer for classification in CNN loses spatial information captured by the convolutional filters from images. Therefore, to preserve the spatial information of image frames, we extracted features from the global average pooling layer on top of the stack of convolutional layers. Hence, the extracted spatial features contain semantically rich visual information from every frame of the video segment with less redundant information.

As shown in Fig. 2 (a), the vital visual features are extracted using pre-trained MobileNet model from each video RGB frame of the given input video segment of size (224 × 224 x 3 × N). Where N is the length of the video segment. After extracting the features from every frame, the features are concatenated to form the feature segment of size (1024 × N). The significance of this process is that the spatial information of each video segment with size (224 × 224 x 3 × N) is reduced to (1024 × N) with meaningful spatial information and less redundancy. Similarly, from the depth video segments, features are extracted from each depth frame and concatenated to form the depth feature segments of size (1024 × N). These RGB and depth feature segments are then concatenated to form the combined feature segment of size (2048 × N). The combined feature segment contains the spatial information of individual frames from both the RGB and depth video segments.

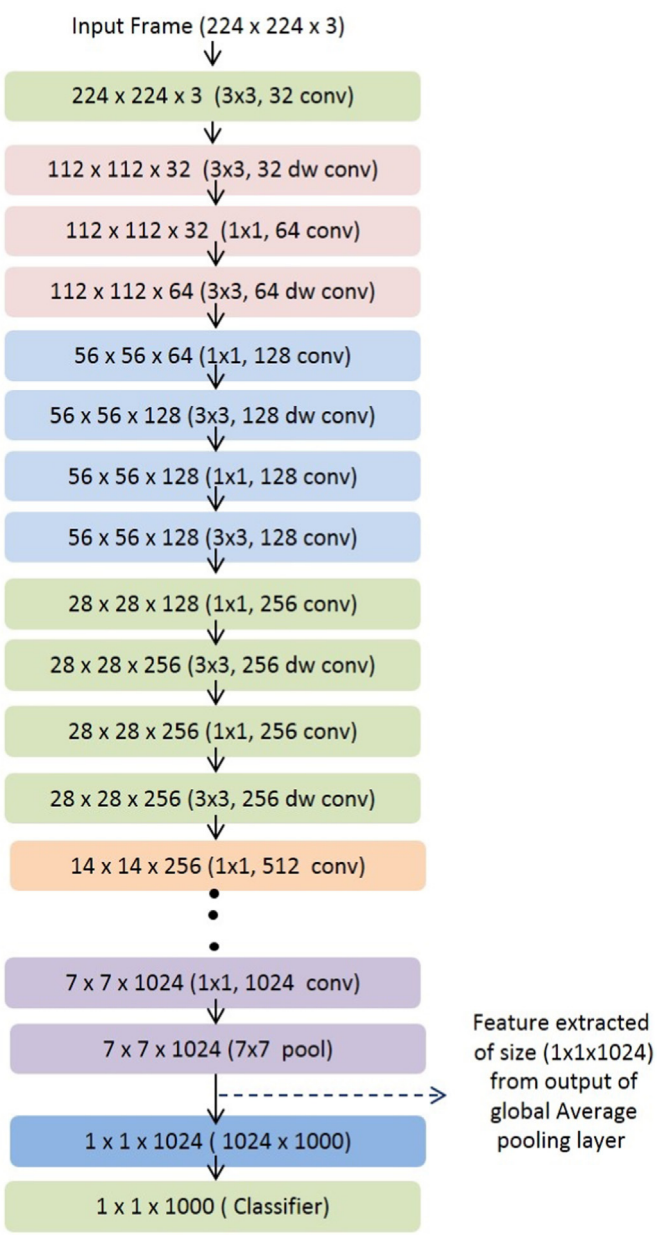

Fig. 3. It shows the layered architecture of MobileNet model (Howard et al., 2017) with the input size and the convolutional filters in the architecture and the global average pooling layer from which the compact features are extracted.

3.2. Bi-LSTM encoder-decoder

The anomaly in a video may occur due to the abnormality in the spatial domain i.e. abnormality in an individual frame and it can also occur in the temporal domain due to the abnormal motion in a sequence of frames. To detect the abnormality in the spatio-temporal domain across frames, the normal pattern sequence of spatial information of frames extracted by the feature extraction module is modeled using a BI-LSTM autoencoder-decoder. The LSTMs are widely known to model the temporal sequence of video, text, audio, and other sequential data and have achieved remarkable success in text recognition (Tai et al., 2015; Bahdanau et al., 2014; Minaee et al., 2021) and speech recognition applications (Graves et al., 2013; Graves and Jaitly, 2014; Alam et al., 2020).

As shown in Fig. 2, the combined feature segment containing the spatial information of RGB and depth segments are given as input to the Bi-LSTM autoencoder-decoder. The autoencoder consists of a two-level hierarchy of Bi-LSTM nodes both at the encoder and decoder level. The first layer of Bi-LSTM with 512 nodes encodes the temporal information of spatial feature segments and the output of the first Bi-LSTM layer is given to the next Bi-LSTM layer of 256 nodes. Similarly, at the decoder level, the inner layer contains 256 Bi-LSTM nodes followed by the 512 nodes at the last Bi-LSTM layer of the decoder. The hierarchy of Bi-LSTM layers at decoder reconstructs the spatial feature segments at the output. The significance of using the Bidirectional-LSTM is that it learns the temporal sequence in both the forward direction and backward direction of the given input sequence. Hence, the normal pattern of sequences can be learned in both directions using the Bi-LSTM layers.

The advantage of using the combined (RGB + Depth) video features is that both the RGB and depth modalities are fused simultaneously which complements each other for learning normal patterns of spatiotemporal sequences. The significance of the proposed semi-supervised approach is that only normal samples are required for training and hence the approach can be applied in real-world settings where the abnormal (unsafe) samples are not easily available for training. The proposed framework is evaluated on the collected RGB+D dataset for ATM surveillance and other real-world benchmark datasets. The results of the proposed framework with quantitative and qualitative evaluation parameters are discussed in the result section 5.

4. Dataset description

This section presents the details of the collected dataset for human actions in ATM named HARD-ATM (Human Action Recognition Dataset in ATM). To discuss the importance of the collected RGB+D dataset in surveillance of ATMs and to differentiate between actions of general gaming/household environment with that

of actions in ATMs, we briefly review some of the available RGB + D datasets for action recognition and surveillance datasets for anomaly detection. Comparison of these datasets with different factors like the number of video samples, data modalities present, subjects involved, intended applications is given in Table 1.

The datasets reviewed in Table 1 consist of general actions like jumping, walking, hand waving, etc., and household activities like drinking, cooking, brushing, and so on. These datasets are created with an intended application to gaming & household purposes and may be less useful to domain-specific activities that are performed in a critical and restricted environment.

Also, most of the available surveillance datasets are collected with a focus on crowd and pedestrian analysis, and very less attention is given to surveillance of indoor scenes. However, as discussed previously actions in ATM are significantly different from gaming and household actions. Therefore, considering all the challenges, sensitiveness, and importance of recognizing actions for surveillance in ATMs, we present a unique RGB + D dataset of human actions in ATM, named as (HARD-ATM). For the collection of the dataset, Microsoft Kinect v2 sensor was used. Three modalities RGB, depth, and skeletal data are captured using this depth sensor. Captured color RGB images has a resolution of 1920 × 1080 pixels. Raw depth images has resolution of 512 × 412 pixels. Both the RGB and depth videos are simultaneously captured at the rate 20 frames per second. The Kinect v2 sensor tracks and captures the skeleton of human being and provides 25 skeleton joints of human structure. Skeleton joint information consists of 3d joint locations of human body in (x, y, z) cartesian coordinate space, where the z coordinate value of a joint, is the distance of that joint from the camera. The camera was mounted on top of the ATM to capture the front view of a human being.

By observing human actions and their behaviors that most frequently occurs in an ATM, we replicated and collected these activities listed in Table 2. We considered a single person as well as multi-person scenarios to capture safe and unsafe activities. In general, 16 activities are collected, and for each activity, three sample instances are recorded. The description of the activities is given in Table 2. The dataset is collected with five different subjects of distinct heights for variations of human activity samples. For each subject, total activity samples are (16 activities x 3 instances) = 48. The average time duration of the activities for each subject is approximately 15 min. The average number of RGB and depth frames present per subject is 18,136 frames. The dataset has total 240 video samples of RGB+D data. Fig. 4 shows the distribution of frames for 240 video samples present in the dataset.

The dataset has 91,460 frames of each RGB and depth videos and the total time duration of the dataset is approximately 1 hr, 15 min. The duration of each activity varies from 300 frames for normal activity of transaction to 1300 frames for the activity of a longer duration. Detailed statistics (activity-wise and subject-wise) of the dataset is given in the supplementary material.

Video samples of the dataset are publicly available at:

https://cse.vnit.ac.in/people/praveenkumar/research/had-atm-4/

Table 2
Activity description of normal and abnormal activities in HARD-ATM dataset, #AL-Activity Label.

AL	Activity Description
A Normal/Safe activities are:	
1)	transaction without wallet
2)	transaction with wallet
3)	transaction with bag
4)	scratching head/face while transacting
5)	card/cash fallen while transacting
6)	card/cash fallen in the pathway
7)	transaction while talking on phone
8)	transaction of longer duration
B Abnormal/Unsafe activities are:	
9)	Moving around and hitting to the ATM
10)	Talking on the phone and moving around.
11)	Moving around and searching (Two person)
12)	Moving around and hitting to the ATM (Two person)
13)	No transaction and talking to each other (Two Person)
14)	Cash/Card stolen from transactor- robbery (Two person)
15)	Cash/Card stolen from transactor- robbery (Three person)
16)	Multi-person Transaction (Two Person)

5. Experimental setup and results

This section discusses the experimental setup and results of the proposed semi-supervised deep learning framework for video anomaly detection. We evaluate the proposed framework on the collected HARD-ATM dataset for ATM surveillance and other challenging video anomaly datasets: Avenue dataset (Lu et al., 2013) and UCF-Crime2Local dataset (Landi et al., 2019). The discussion starts with the experimental setting for training and evaluation on HARD-ATM dataset followed by the results on Avenue, and UCF-Crime2Local dataset.

5.1. Experimental setup

To match with the input layer of MobileNet, each video frame is resized to 224 × 224 × 3. The size of video segment N is fixed to 10 frames. For training, only the normal video samples are used. The 5-fold cross validation protocol is followed i.e. normal samples from 4-subjects (96 video samples) are used for training and remaining 144 video samples are given for testing. This is done for each cross-subject validation setting. The Bi-LSTM autoencoder is trained with a learning rate of 0.001 and batch size of 32. A dropout rate of 0.2 is added between LSTM layers to avoid over-fitting. For quantitative and qualitative evaluation of the proposed framework, we consider *precision*, *recall* and $f_{measure}$ as evaluation parameters. Following paragraph discuss the notations and meanings to define true positive, true negative, false positive, and false negative.

Table 1
Review of RGB + D action and surveillance datasets with applications. Abbreviations used #A- Number of action classes, #ST- Total number of subjects involved, #S-Total number of samples present, M- Modality present, (S/M)- Single or multiple person involved.

Dataset	#Frames	#A	#ST	#S	M	S/M	Applications
UCSD pedestrain-1 (Mahadevan et al., 2010)	1400	–	–	70	RGB	M	Pedestrian-Surveillance
UCSD pedestrain-2 (Mahadevan et al., 2010)	4560	–	–	28	RGB	M	Pedestrian-Surveillance
Avenue (Lu et al., 2013)	30,652	–	–	37	RGB	M	Pedestrian-Surveillance
CAD-60 (Sung et al., 2012)	43,200	12	4	60	RGB + D	S	Household/office
CAD 120 (Koppula et al., 2013)	61,585	10	4	120	RGB + D	S	Household/office
Subway-exit (Adam et al., 2008)	72,401	–	–	–	RGB	M	Pedestrian-Surveillance
HARD-ATM (Ours)	91,460	16	5	240	RGB + D	S/M	ATM-Surveillance

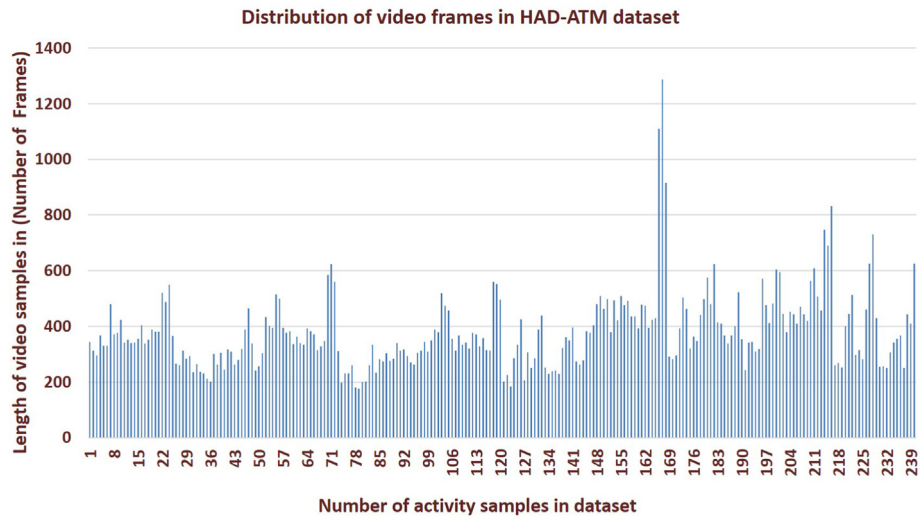

Fig. 4. Distribution of the video frames present in the HARD-ATM dataset.

★True Positives (Tp): abnormal activity sample detected as anomalous
★True Negatives (Tn): normal activity sample detected as non-anomalous
★False Positives (Fp): normal activity sample detected as anomalous
★False Negatives (Fn): abnormal activity sample detected as non-anomalous

Precision (P) is the fraction of the actual abnormal activity sample detected as anomalous among the identified activities detected as anomalous. Recall (R) is the fraction of actual abnormal activity sample detected as anomalous among total abnormal activity samples that are present.

$$Precision = \frac{Tp}{(Tp + Fp)} Recall = \frac{Tp}{(Tp + Fn)}$$

$F_{measure}$ is generally used as parameter to test the performance of binary classification model. It is the harmonic average of the precision and recall. Value of F1 score lies in closed interval [01]. Where value 1 indicates perfect classification and 0 means worst classification.

$$F_{Measure} = \frac{2 \cdot Precision \cdot Recall}{(Precision + Recall)}$$

Overall accuracy of the binary classification for detection of anomalous sample can be calculated as given below:

$$Accuracy = \frac{Tp + Tn}{(Tp + Tn + Fp + Fn)}$$

5.2. Results

As discussed in related work section, there are some work on detection of abnormal activity in ATM environment reported in Liu et al. (2012); Nar et al. (2016); Lee et al. (2018). However, there are no qualitative and quantitative results reported in these papers. Also, the absence of data/dataset used for experimentation in their work and the lack of results with qualitative evaluation parameters makes it difficult for us to evaluate and compare our approach with them. However, this section discusses a detailed statistical result analysis of our proposed semi-supervised approach on HARD-ATM dataset.

For testing, a sample video is given as input to the encoder and the reconstruction error between the input video features and the reconstructed output features is calculated. If the error is above a set threshold, then the given test video input can be detected as anomalous which deviates from the normal ones. Here, we set the threshold as the *mean* + 0.5std of the total reconstruction error calculated for the given training samples. Fig. 5 shows the result of the proposed Bi-LSTM autoencoder trained using combine video features of RGB and depth data. It shows the reconstruction error for all the 5- fold cross-validation settings. The figure shows the graph of the reconstruction errors obtained for individual RGB, depth, and the combined RGB + depth features. Each fold consists of 144 test samples with 24 normal samples and the remaining 120 abnormal/anomalous samples. The green label in graph represents the normal and the red label represents the abnormal samples. For instance, Fold-1 consists of 24 normal samples and 120 abnormal samples and out of the total 144 video samples, the proposed approach using the combined RGB + depth features recognizes 34 samples as normal video samples and 110 as abnormal samples with a *precision* of 0.990, *recall* of 0.908 and $f_{measure}$ of 0.947.

Table 3 shows the results of each fold of 5-Fold cross-validation setting for detecting the anomalous samples. It shows the comparison of results for the individual modality of RGB, depth data, and the combined (RGB + Depth) features. From table it can be observed that, with individual RGB video features, the framework is able to detect abnormal activity with *precision* of 0.935 and $f_{measure}$ of 0.928. For individual video features of depth modality, the proposed framework detects the anomaly with 0.943 *precision*. On combining the RGB and depth spatio-temporal features, the accuracy of anomaly detection increases with *precision*, *recall* and $f_{measure}$ of 0.963, 0.938 and 0.949 respectively. By analyzing the results, it can be seen that combining the RGB and depth video segment features improves the accuracy for detection of anomalous activities. The advantage of the proposed unsupervised approach is that only normal samples are required for training and hence the approach can be applied in real-world settings where the abnormal/anomalous samples are not easily available for training. To justify the effectiveness of the proposed approach in other real-world environments, we also evaluated the proposed framework on other benchmark video anomaly detection datasets: Avenue and

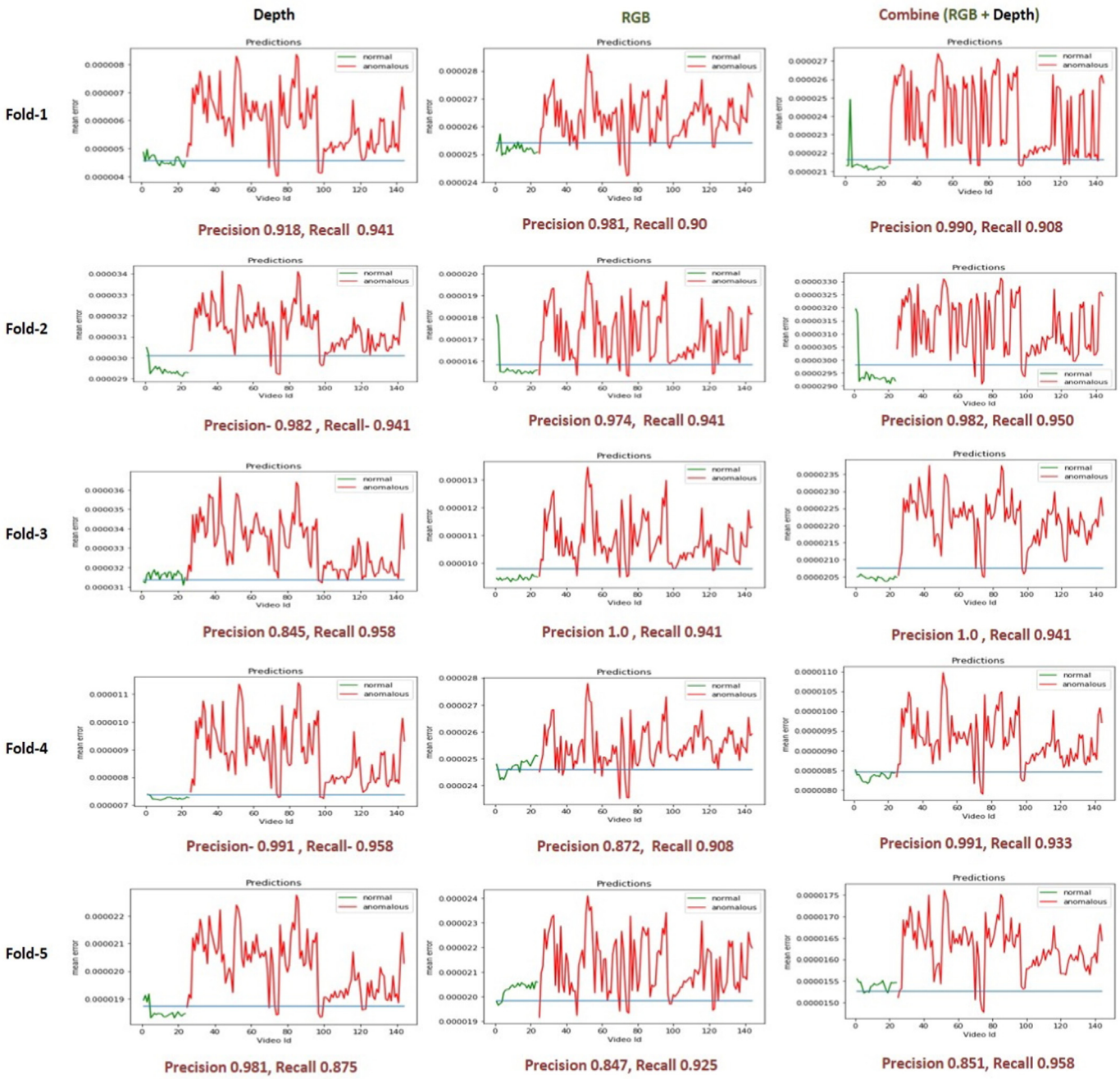

Fig. 5. Result of the proposed Bi-LSTM autoencoder trained using Depth, RGB, and combined (RGB + depth) video features. It shows the performance and reconstruction error for all the 5- fold cross-validation settings on individual RGB, depth, and combined (RGB + depth) modality. There are 144 test video samples present in each fold with 24 normal samples and 120 abnormal video samples. The green label represents the mean reconstruction error generated for the normal video samples and the red label represents the mean reconstruction error generated for abnormal samples.

UCF-Crime2Local. The following paragraphs discuss the results on Avenue and UCF-Crime2Local datasets.

The Avenue dataset was collected by Lu et al. (2013) and consists of normal and abnormal video samples to detect the anomaly at the railway station platform. The dataset has 16 normal videos for training and 21 anomalous video samples for testing. Some of the anomalies in the dataset are running at the platform, throwing bag/papers at the platform, etc. To train the proposed framework, only normal samples of the dataset are used. For training and testing, only RGB modality is utilized, as the dataset contains only RGB videos.

Fig. 6 shows the reconstruction error for the test anomalous video sample of the person running at the railway station platform. It shows the mean reconstruction error for each video segment of the video sample. The video sample has total 947 RGB frames and divided into video segments with each segment size fixed to 10 frames. The plot shows the video segment id on the x-axis and the reconstruction error is plotted on the y-axis. It also shows the marked video segments with their mean reconstruction error and the corresponding RGB frames. From the figure, it can be observed that the video segments having the anomalous behavior of person running at the platform has the high reconstruction error and

Table 3
Results of each fold of 5-Fold cross-validation setting for detecting anomaly detection using proposed semi-supervised approach. It also shows the comparison of individual of RGB, depth modalities and the combine (RGB + Depth) modality features.

RGB						
Evaluation parameter	F-1	F-2	F-3	F-4	F-5	Avg
Precision	0.981	0.974	1.0	0.872	0.847	0.935
Recall	0.90	0.941	0.941	0.908	0.925	0.923
F-measure	0.939	0.957	0.970	0.890	0.884	0.928
Depth						
Evaluation parameter	F-1	F-2	F-3	F-4	F-5	Avg
Precision	0.918	0.982	0.845	0.991	0.981	0.943
Recall	0.941	0.941	0.958	0.958	0.875	0.935
F-measure	0.929	0.961	0.898	0.974	0.925	0.938
Combine-RGB + Depth						
Evaluation parameter	F-1	F-2	F-3	F-4	F-5	Avg
Precision	0.990	0.982	1.0	0.991	0.851	0.963
Recall	0.908	0.950	0.941	0.933	0.958	0.938
F-measure	0.947	0.966	0.970	0.961	0.901	0.949

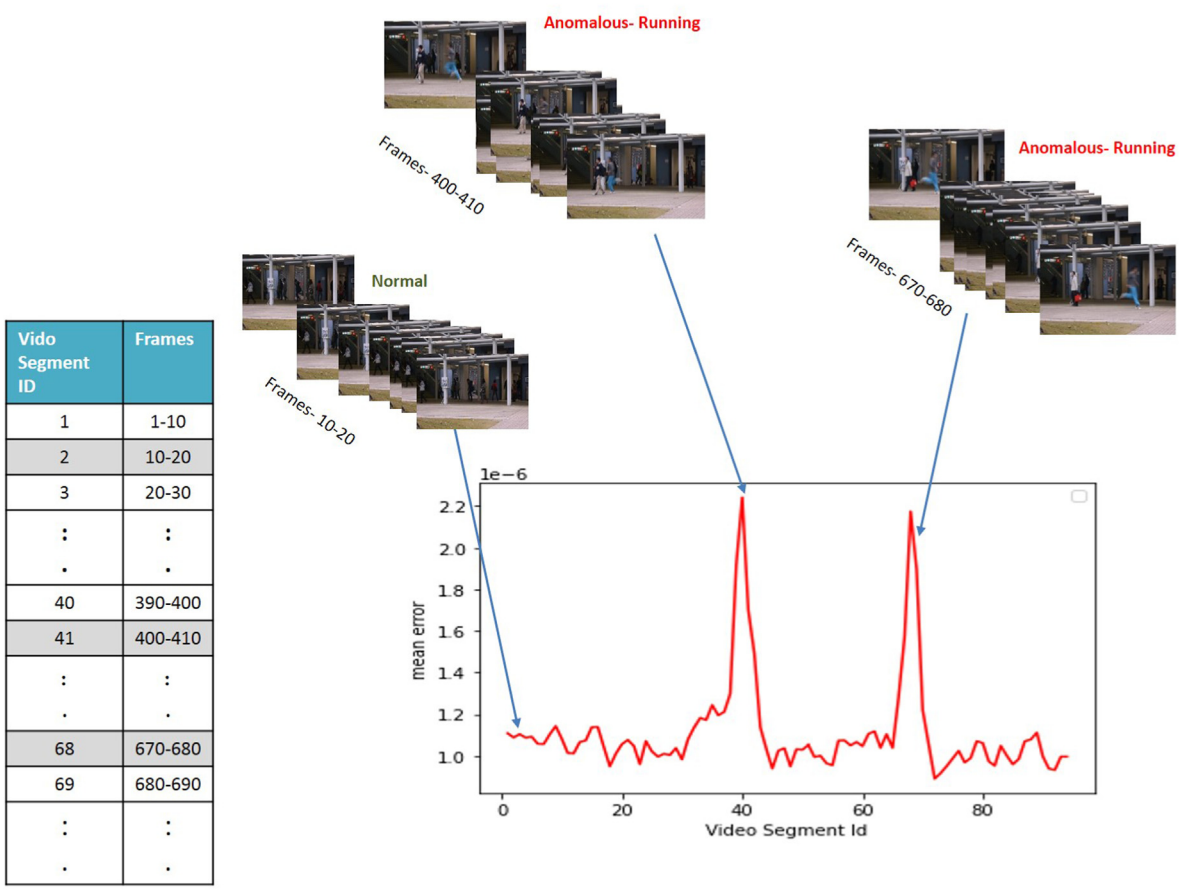

Fig. 6. It shows the mean reconstruction error generated by the proposed approach for the anomalous test video sample-4 of the person running at the railway station platform from Avenue Dataset (Lu et al., 2013). It shows the mean reconstruction error for each video segment of the video sample. The marked points on the plot show the mean reconstruction errors for the video segments- 2 (Normal), 41 and 68 (Anomalous-Running) with their corresponding RGB frames.

hence detected as the anomalous. The proposed semi-supervised framework detects the anomalous samples in the dataset with an accuracy of 89.1%.

Table 4 shows the performance and comparison of the proposed method with other state-of-the-art methods. Although the accuracy achieved by state-of-the-art (Lee et al., 2020) is slightly better than our proposed approach. The framework adopted in (Lee et al., 2020) requires the reconstruction of raw input frames at the decoder output. In addition, the framework also constructs the attention maps from the raw input frames that add to the complexity. The main advantage of our proposed approach over (Lee et al., 2020) is that our approach does not require the overhead of reconstruction of the raw frames at the decoder output and it only requires reconstruction of compact features at the decoder

Table 4
Comparison of the results of proposed approach on Avenue dataset with other state-of-the-art methods.

Method	Accuracy (%)
Hasan et al. (2016)	70.2
Luo et al. (2017)	77.0
Ribeiro et al. (2018)	77.2
Chong and Tay (2017)	80.3
Zhao et al. (2017)	80.9
Smeureanu et al. (2017)	84.6
Tang et al. (2020)	85.1
Wang et al. (2020)	87.0
Lee et al. (2020)	90.0
Proposed	**89.1**

Table 5
Comparison of the results of proposed approach on UCF-Crime2Local dataset with other state-of-the-art methods.

Method	Accuracy (%)
Gianchandani et al. (2019)	68.0
Landi et al. (2019)	74.73
Weakly supervised Landi et al. (2019)	80.96
Ullah et al. (2020)	89.05
Proposed	**91.1**

output that makes it much simpler and computationally efficient. Therefore, our proposed approach can be used in real-time embedded systems with low computational requirements for the detection of anomalies.

The UCF-Crime2Local dataset was prepared by (Landi et al., 2019). The dataset is the subset of the UCF-Crime dataset collected by (Sultani et al., 2018). The UCF-Crime2Local dataset contains total 300 RGB video samples with 200 normal and 100 abnormal video samples. The dataset contains video samples from real-world environments like shops, streets, etc. It contains six anomalous human activities like assault, robbery, and so on.

To evaluate the proposed framework we used the training and testing splits given in (Landi et al., 2019). For training, only RGB modality is used as the dataset contains only RGB video samples. The train and test splits in the dataset contain both the normal and anomalous samples for supervised training. However, for the training of our proposed semi-supervised framework, we used only normal video samples present in the dataset and tested the proposed framework on the test video samples experimented in (Landi et al., 2019).

Table 5 shows the performance of the proposed semi-supervised CNN−Bi-LSTM framework on UCF-Crime2Local dataset. It shows the performance of the proposed approach with other recent methods applied for video anomaly detection. From Table 4 and Table 5, it can be observed that the proposed framework has achieved impressive results on Avenue and UCF-Crime2Local datasets respectively for video anomaly detection in the outdoor scenarios. Moreover, the proposed framework outperforms the previous methods for the classification of anomalous samples in UCF-Crime2Local dataset. Hence, results on HARD-ATM, Avenue, and UCF-Crime2Local datasets show that the framework can be applied in both the critical indoor environments like ATMs as well as the real-world outdoor environments like railway platforms, shops etc. for the detection of any anomalous event.

6. Conclusion and future scope

Automatic video anomaly detection has become very important in the modern era of smart cities for surveillance and security of private and public places. In this paper, a multi-modal semi-supervised CNN-BiLSTM based framework is presented for the detection of anomalous event in the critical environments like Bank-ATMs. The proposed framework utilizes the combined features of RGB and Depth modalities from advanced RGB+D sensor for anomaly detection. The advantage of the approach is that only weakly labeled normal video samples are required for training. Also, to reduce the computational complexity for training and detection, rich visual features were extracted from raw RGB and Depth frames. Moreover, due to the unavailability of any dataset for ATM surveillance in the public domain, a unique RGB+D dataset for surveillance of ATM is contributed in this paper. Furthermore, the impressive results of the proposed framework on other real-world benchmark datasets show that it can be applied in both indoor and outdoor real-world environments for anomaly detection.

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